

Cooperative Air-Ground Robot Localization

- final project description

AEM 5451

1 Introduction

There are several applications where an unmanned ground vehicle would need to operate in locations where GPS is challenged or denied (urban navigation). In those same applications, it would be necessary for that ground vehicle to have accurate estimates of its position. In this project, a cooperative localization problem is setup to leverage the use of an unmanned aerial vehicle to operate in conjunction with the ground vehicle.

2 Physical system

The system consists of a 4-wheeled unmanned ground vehicle (UGV) that moves along the ground while cooperatively tracking and communicating with an unmanned aerial vehicle (UAV) that flies a 2D path at a constant altitude above the ground vehicle. The UAV and UGV are controlled by independent inputs, but obtain relative measurements to one another through their encounter. The UGV cannot obtain reliable GPS measurements and effectively switches its receiver off during its encounter with the UAV, while the UAV maintains reliable GPS throughout the encounter. This assignment will focus on the development of the centralized estimator only, using simple non-linear vehicle motion models.

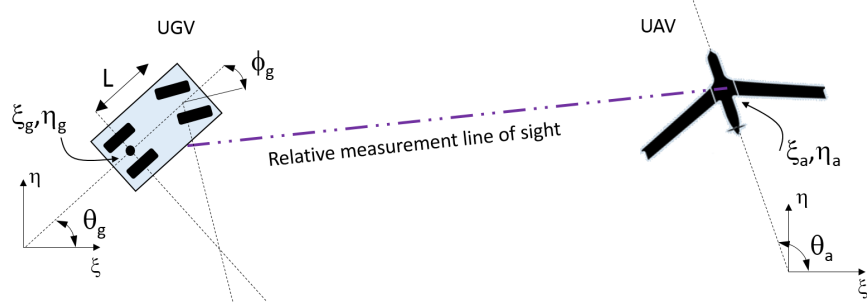


Figure 1: UAV-UGV cooperative localization problem setup.

3 Dynamical system

The UGV's motion is modeled kinematically here as a simple 4-wheeled steerable Dubin's car with front-rear wheel separation length L , with East position ξ_g and North position η_g in an inertial frame, and with heading angle θ_g . The UGV's control inputs are $\mathbf{u}_g = [v_g, \phi_g]^T$, the linear velocity (m/s) and steering angle (rad), respectively. This leads to the non-linear equations of motion

$$\dot{\xi}_g = v_g \cos \theta_g + \tilde{w}_{x,g} \quad (1)$$

$$\dot{\eta}_g = v_g \sin \theta_g + \tilde{w}_{y,g} \quad (2)$$

$$\dot{\theta}_g = \frac{v_g}{L} \tan \phi_g + \tilde{w}_{\omega,g}, \quad (3)$$

where $\tilde{\mathbf{w}}_g(t) = [\tilde{w}_{x,g}, \tilde{w}_{y,g}, \tilde{w}_{\omega,g}]^T$ is the process noise on the UGV states.

The fixed-wing UAV motion is likewise modeled kinematically as a simple Dubin's unicycle, with East position ξ_a and North position η_a in an inertial frame, with heading angle θ_a . The UAV's control inputs are $\mathbf{u}_a = [v_a, \omega_a]^T$, the linear velocity (m/s) and angular rate (rad/s), respectively. This leads to the non-linear equations of motion

$$\dot{\xi}_a = v_a \cos \theta_a + \tilde{w}_{x,a} \quad (4)$$

$$\dot{\eta}_a = v_a \sin \theta_a + \tilde{w}_{y,a} \quad (5)$$

$$\dot{\theta}_a = \omega_a + \tilde{w}_{\omega,a} \quad (6)$$

where $\tilde{\mathbf{w}}_a(t) = [\tilde{w}_{x,a}, \tilde{w}_{y,a}, \tilde{w}_{\omega,a}]^T$ is the process noise on the UAV states.

Hence, the combined system state, control inputs, and disturbance inputs are

$$\begin{aligned}\mathbf{x}(t) &= [\xi_g \quad \eta_g \quad \theta_g \quad \xi_a \quad \eta_a \quad \theta_a]^T, \\ \mathbf{u}(t) &= [\mathbf{u}_g \quad \mathbf{u}_a]^T, \\ \tilde{\mathbf{w}}(t) &= [\tilde{\mathbf{w}}_g \quad \tilde{\mathbf{w}}_a]^T\end{aligned}$$

The sensing model for this system is given by a combination of noisy ranges and azimuth angles of the UGV relative to the UAV, noisy azimuth angles of the UAV relative to the UGV, and noisy UAV GPS measurements,

$$\mathbf{y}(t) = \begin{bmatrix} \tan^{-1} \left(\frac{\eta_a - \eta_g}{\xi_a - \xi_g} \right) - \theta_g \\ \sqrt{(\xi_g - \xi_a)^2 + (\eta_g - \eta_a)^2} \\ \tan^{-1} \left(\frac{\eta_g - \eta_a}{\xi_g - \xi_a} \right) - \theta_a \\ \xi_a \\ \eta_a \end{bmatrix} + \tilde{\mathbf{v}}(t),$$

where $\tilde{\mathbf{v}}(t) \in \mathbb{R}^5$ is the sensor error vector (which can be modeled by AWGN).

4 Nominal system parameters

For this project, assume $L = 0.5$ m for the UGV and that the UGV has a limited range of steering angles $\phi_g \in [-\frac{5\pi}{12}, \frac{5\pi}{12}]$ rads, with a top ground speed of $v_{g,max} = 3$ m/s. Further assume that the the UAV's turn rate is limited to $\omega_g \in [-\frac{\pi}{6}, \frac{\pi}{6}]$ rad/s and flies with velocity $v_a \in [10, 20]$ m/s.

As a nominal trajectory for the UGV, assume that the UGV starts at $(\xi_g, \eta_g) = (10, 0)$ with $\theta_g = \pi/2$ rad, and initiates a steady continuous turning maneuver with $v_g = 2$ m/s and $\phi = -\pi/18$ rad. As a nominal trajectory for the UGV, assume the UAV starts at $(\xi_a, \eta_a) = (-60, 0)$ with $\theta_a = -\pi/2$ rad, and initiates a steady continuous turning maneuver with $v_a = 12$ m/s and $\omega_a = \pi/25$ rad/s.